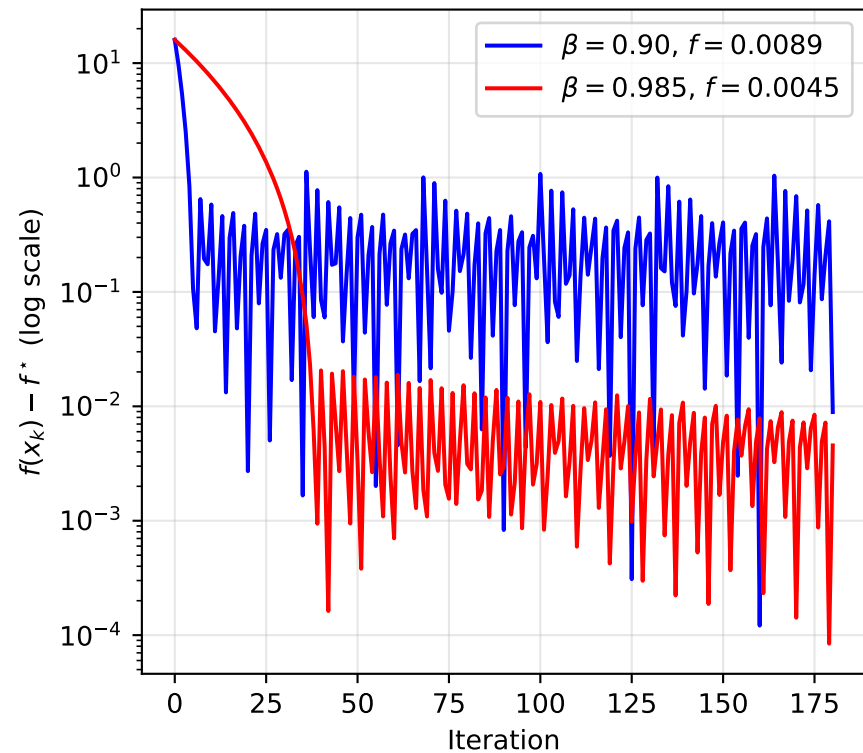


Frank-Wolfe Algorithm Analysis

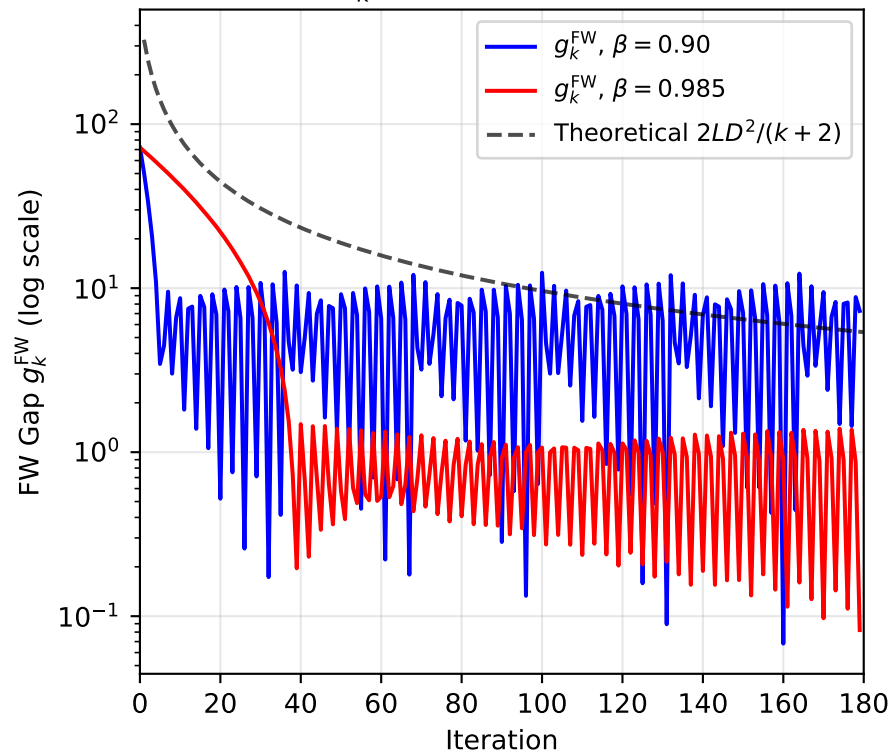
Frank-Wolfe: Interior Optimum

$$f(x) = (x_1 - 1)^2 + (x_2 - 5)^2$$



Frank-Wolfe Optimality Gap

$$g_k^{\text{FW}} = \nabla f(x_k)^T (x_k - z_k)$$



FW Gap as Optimality Certificate

$$f(x_k) - f^* \leq g_k^{\text{FW}}$$

